

Aditya Gunturu

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EDUCATION

- University of Washington, PhD** 09/2026 – Present | Seattle, WA
Computer Science & Engineering
Paul G. Allen School of Computer Science & Engineering
Advisor: Jon Froehlich
- University of Calgary, Master of Science** 08/2023 – 01/2026 | Calgary, Canada
Computational Media Design
Advisor: Ryo Suzuki
- Manipal Institute of Technology, Bachelor of Technology** 07/2018 – 08/2022 | Manipal, India
Major: Media Technology
Minor: Computational Mathematics

PUBLICATIONS

- [C6] **Aditya Gunturu**, Mackenzie Leake, Ryo Suzuki, Stephen DiVerdi. CARVid: Content-Aware Retargeting of Videos with On-Screen Graphics. In Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology (**UIST '26**)
[Preview](#) | [Project Page](#)
Intern project at Adobe Research
- [C5] **Aditya Gunturu**, Ben Pearman, Keiichi Ihara, Morteza Faraji, Bryan Wang, Rubaiat Habib Kazi, and Ryo Suzuki. MapStory: Prototyping Editable Map Animations with LLM Agents. In Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology (**UIST '25**)
[Paper](#) | [Video](#) | [Preview](#) | [Project Page](#)
- [C4] **Aditya Gunturu**, Shivesh Jadon, Nandi Zhang, Jarin Thundathil, Wesley Willett, and Ryo Suzuki. RealitySummary: On-Demand Mixed Reality Document Enhancement using Large Language Models. In *Proceedings of the 2025 ACM Symposium on Spatial User Interaction (SUI '25)*.
[Paper](#) | [Video](#) | [Project Page](#)
- [C3] **Aditya Gunturu**, Yi Wen, Jarin Thundathil, Nandi Zhang, Rubaiat Habib Kazi, and Ryo Suzuki. Augmented Physics: Creating Interactive and Embedded Physics Simulations from Static Textbook Diagrams. In Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology (**UIST '24**) 🏆 **Best Paper Award** | In collaboration with *Adobe Research*
[Paper](#) | [Video](#) | [Teaser](#) | [Project Page](#)
- [C2] Ada Yi Zhao, **Aditya Gunturu**, Ellen Yi-Luen Do, and Ryo Suzuki. Guided Reality: Generating Visually-Enriched AR Task Guidance with LLMs and Vision Models. In Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology (**UIST '25**)
[Paper](#)
- [C1] Haijun Xia, Tony Wang, **Aditya Gunturu**, Peiling Jiang, William Duan, and Xiaoshuo Yao. CrossTalk: Intelligent Substrates for Language-Oriented Interaction in Video-Based Communication and Collaboration. In Proceedings of the Annual ACM Symposium on User Interface Software and Technology (**UIST '23**)

WORK EXPERIENCE

Adobe Research, *Research Scientist Intern*

05/2025 – 08/2025 | San Francisco

Mentors: **Stephen DiVerdi** and **Mackenzie Leake**

- Developed CARVid, an automatic, content-aware retargeting technique for videos with on-screen graphics (lower thirds, title cards, scoreboards) that resizes graphics while preserving readability and original design intent across aspect ratios.
- CARVid combines text-aware scaling, temporally stable content-aware space removal (our improved version of seam carving), and text/layout reflow, selecting operations based on the presence of text and design details within the on-screen graphic.
- Conducted a technical evaluation on a custom dataset of 40 motion-graphics videos, achieving 77% success and interviews with four professional motion-graphics designers described results as close to production-ready for downstream editing.

Interactions Lab, UCalgary, *Graduate Researcher*

08/2023 – 04/2025 | Calgary, Canada

- My research focuses on creating ML-integrated authoring tools that lower the bar for both creators and novices to produce dynamic, animated, and interactive visuals on-demand and in real time. My ongoing project focuses on a simple and intelligent authoring tool for creating motion graphics.
- My work so far has led to two papers at UIST '25, one at SUI '25 and at UIST '24, which also won the best paper award!

Subtl.AI, CIE, *Software Development Intern*

02/2023 – 07/2023 | IIIT Hyderabad

- My work included full-stack product engineering and NLP Research for large-scale document question answering.
- I was on the founder's team completely redesigning their early prototype and implementing it into a scaled product, which is now being used by the biggest bank in India (SBI)
- Built the web-based chatbot interface with various novel interactions for clients to gain insights from their proprietary documents. I also improved the query-document similarity algorithm, which is based on an in-house BERT-based model. I also contributed to integrating an open-source LLM into this pipeline.

Empathic Computing Lab [✉](#), *Research Intern/Collaboration (Remote)*

09/2022 – 01/2023 | Remote

Under Prof. Mark Billinghurst

I built a cross-device collaboration system between a mobile AR device and a remote desktop collaborator, where a desktop client can drop and manipulate objects in the remote AR scene in real-time. I used React and WebRTC for the desktop interface and Unity for the AR system. I worked with various AR frameworks like Niantic Lightship and Google's geospatial API to explore multiple interactions. Firebase's real-time database was used for seamless data integration.

University of California, San Diego [✉](#)

06/2021 – 09/2022 | Remote

Research Intern/Collaboration (Remote)

*At the **Creativity Lab** [✉](#), Under Prof. Haijun Xia*

- I built a context-aware intelligent video conferencing platform that processes language from conversations into action and content recommendations in real-time.
- I was responsible for implementing the system interactions for action/content recommendations and used webRTC, firebase real-time database, and React (Typescript) to build the prototype. Several custom components were built to set up real-time panel communication for the front-end, NLP Module, and user interactions.

- I also worked on a part of the NLP module. Specifically, I built the action recommendation module using BERT (sBERT) and spaCy. I devised a novel method that calculates sentence similarity between a custom corpus and live transcripts to recommend relevant UI actions in real time. *Our paper is **published at UIST '23!***

FliteX, MITACS Accelerate Intern

01/2024 – 08/2024

Built a web-based interactive 3D flight path simulator for Transatlantic flights.

As part of the MITACS accelerate fellowship program, which allows students to work on funded industry projects.

NOTABLE AWARDS

- Best Paper Award, ACM UIST 2024**
- 2024
- Augmented Physics got awarded the Best Paper at UIST 2024! The award is presented to the top 1% of accepted papers.
- Alberta Graduate Excellence Award, \$11,000**
- 2024
- Awarded \$11,000 for research and scholarly excellence by the province of Alberta.
- MITACS Accelerate Fellowship, \$20,000**
- 2024
- Funding provided by Mitacs for conducting research with an industry partner.
- BIRAC - E-YUVA Entrepreneurship Grant** [↗](#), \$10,000
- 2022
- As part of the Flexible Electronics lab under the startup incubator, our group (three students, including myself) won an award for undergraduate entrepreneurship efforts. The funding enabled the lab to further develop its technology.

SERVICE

- DIS 2025, Reviewer**
- 03/2025 – 03/2025
- CHI LBW 2025, Reviewer**
- 02/2025 – 02/2025
- CHI 2025, Reviewer**
- 11/2024 – 11/2024
- CHI 2026, Reviewer**
- 09/2025 – 12/2025
- UIST 2025, Reviewer**
- 04/2025 – 07/2025

PRESS COVERAGE

- New interactive AI tool makes textbooks come to life** [↗](#), *BGR*
- Creating Interactive and Embedded Physics Simulations from Static Textbooks** [↗](#), *Hacker News*
- Interactive Physics: A New Tool for Students** [↗](#), *AZO Robotics*
- Textbooks come alive with new AI tool** [↗](#), *Phys.org*
- Science textbooks come alive with new AI tool** [↗](#), *The Brighter Side News*
- Textbooks Come Alive with 'Augmented Physics' – Transforming Education with AI** [↗](#), *Opentools.ai*